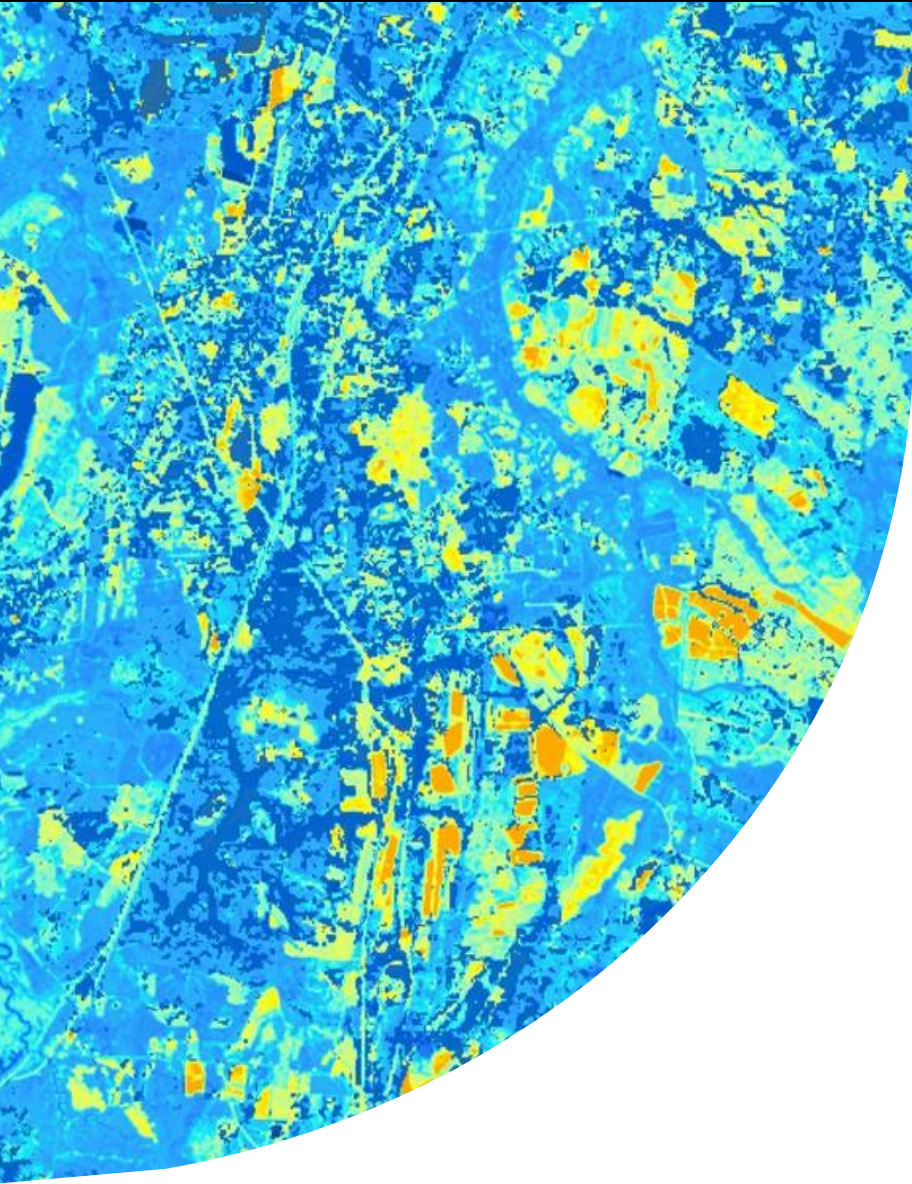




National Aeronautics and
Space Administration



Coastal South Carolina Water Resources II

Assessing Unprotected Wetlands to Identify
Priority Conservation Areas for Community
Protection Against Flood Events in South
Carolina

Asa Julien, Jake Ferus, Karen Wang, & Sydney
Brown (Analytical Mechanics Associates)

Georgia – Athens | Summer 2025



Meet the Team



Asa Julien



Jake Ferus



Karen Wang



Sydney Brown



Wetlands

Wetlands are diverse ecosystems found at the transition between aquatic and terrestrial environments. The ground is either permanently or seasonally saturated with water.



Image Credit: National Wildlife Refuge Association

An **isolated wetland** is a wetland that is not directly connected to a navigable water of the United States **via continuous surface connection**.



Coastal South Carolina Water Resources II



Image Credit: Karen Wang

Provide Habitat



Image Credit: Uwe Aranas

Sequester Carbon



Image Credit: Billy Wilson

Mitigate Floods



Image Credit: The Coastal Conservation League

Wetlands and Flood Impacts

Wetlands act as a **natural sponge**, storing water during wet periods, and then releasing it during dry periods.

A wetland can store around **one million gallons** of water per acre.



Wetland Degradation

The long-term global loss of naturally occurring wetlands has been estimated to be between **54-57%**, with the rate of loss being **3.7x faster** in the 20th and 21st centuries.

The United States has lost an estimated **40%** of its inland wetland coverage since 1700.

The 2023 Sackett v. EPA ruling excludes isolated wetlands from federal protection.

Sackett v. EPA

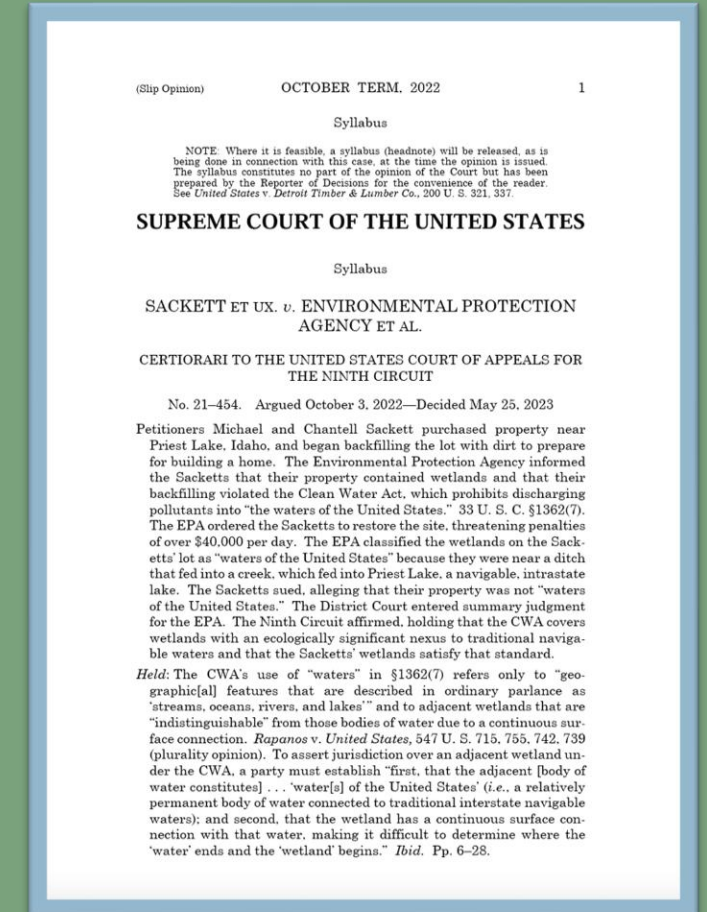


Image Credit: The United States Supreme Court



The Coastal Conservation League

The **Coastal Conservation League** is a non-profit organization that promotes improved sustainability initiatives and aims to protect natural resources of South Carolina.

They work with different state agencies, local businesses, and citizen groups.

Currently, they work to preserve the natural landscape and assess social vulnerability impacts from natural disasters by **focusing on wetlands and their benefits to communities.**





Community Concerns

Isolated wetlands are **not being protected by federal law**



Potential decrease of wetlands causes an **increase in risk of flooding severity**



These flooding events negatively **impact certain communities**

Image Credit: The Coastal Conservation League



Term I Project

The 2025 **Spring** GA Coastal South Carolina Water Resources Team mapped isolated wetlands in **Jasper, Berkeley, and Horry** South Carolina that have lost protection. They also mapped the decrease in wetlands from **2015-2025**.

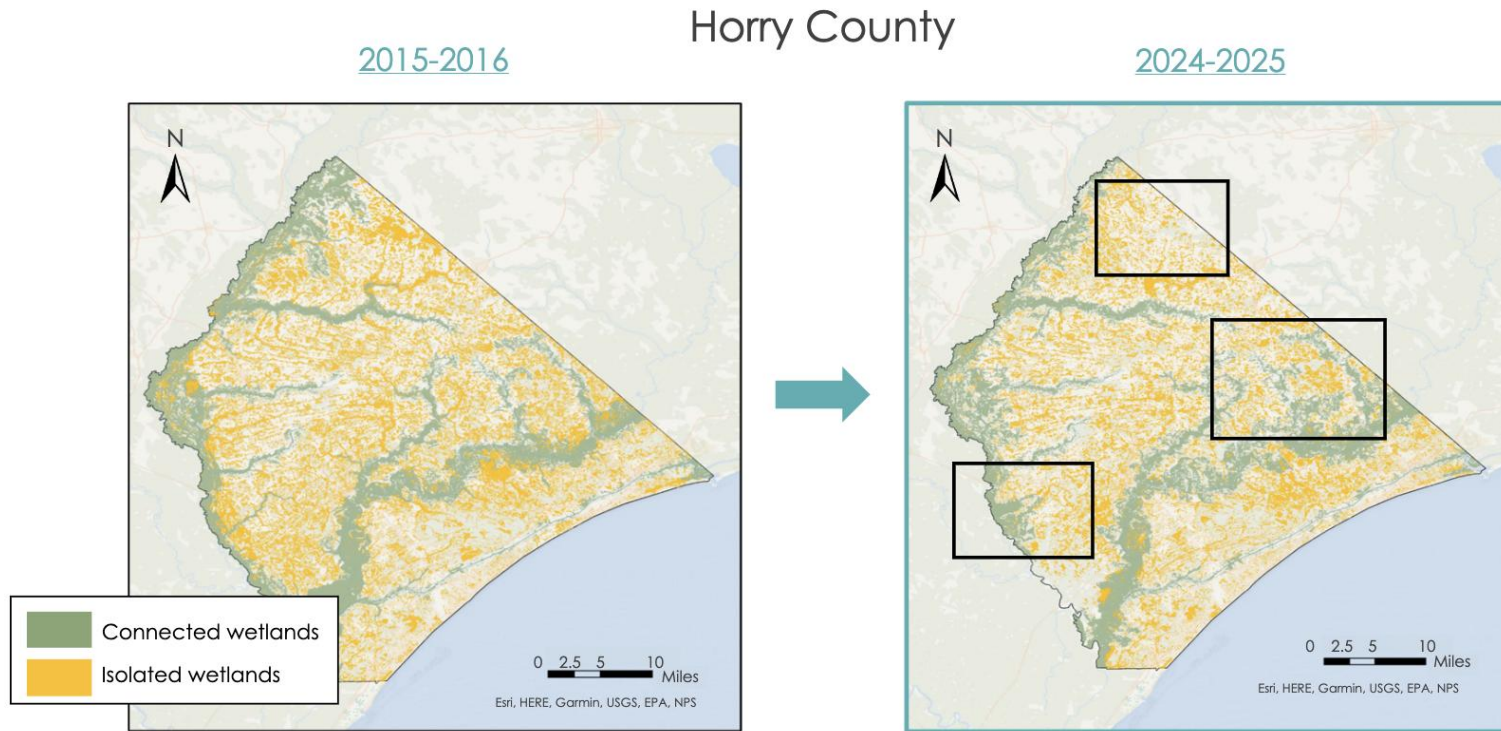


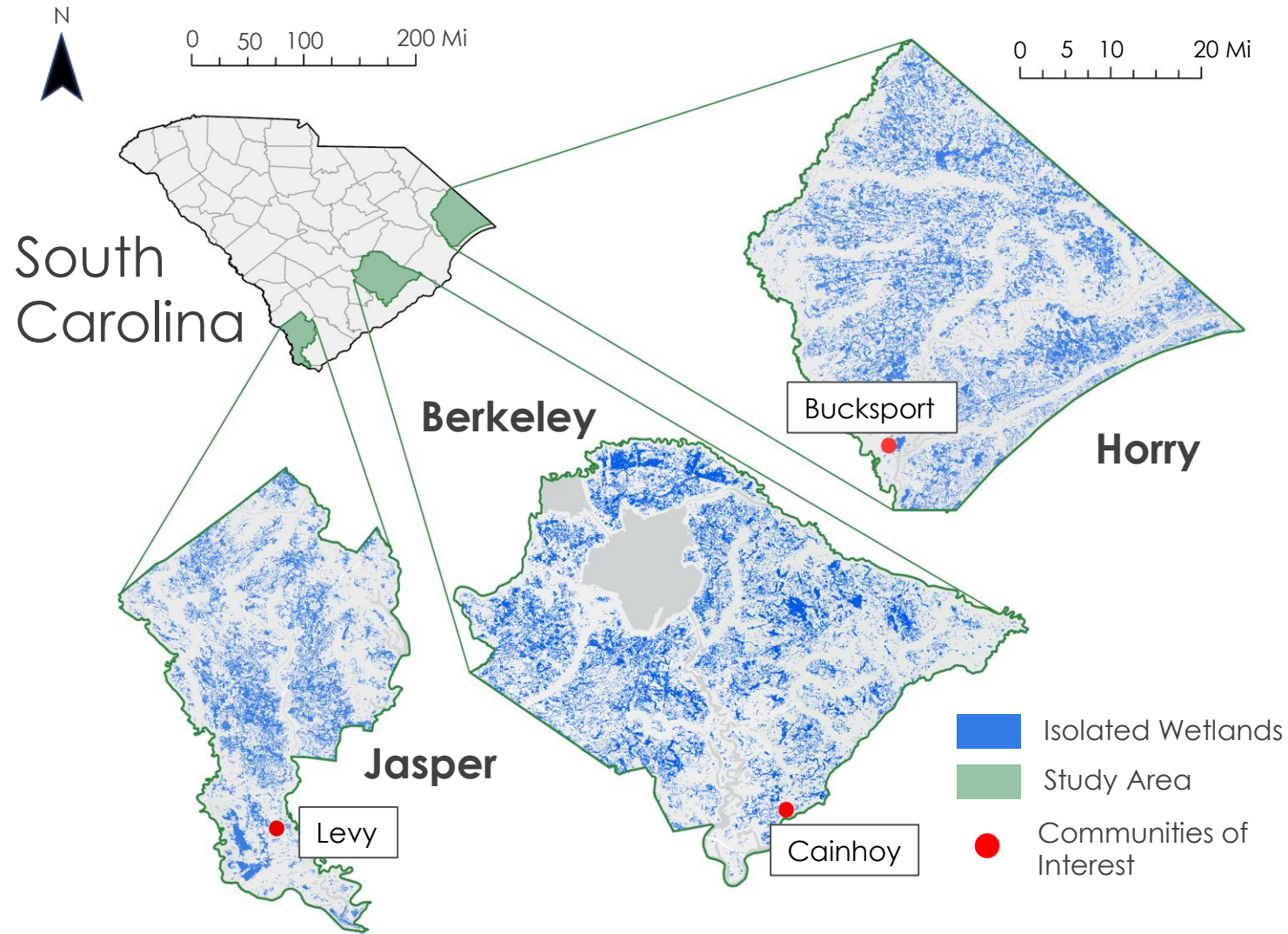
Image Credits: 2025 Spring GA Coastal South Carolina Water Resources Team



Image Credit: 2025 Spring GA Coastal South Carolina Water Resources Team

**2025 Spring GA Coastal
South Carolina Water
Resources Team**

Study Area & Time Period



Sources: ESRI, TomTom, Garmin, FAO, NOAA, USGS, © OpenStreetMap contributors, and the GIS User Community



Study Area

- Jasper, Berkeley, & Horry Counties
- Levy, Cainhoy, & Bucksport Communities

Study Period

- 2015 – 2025
- 2018 – **Hurricane Florence**



Image Credit: NASA

Project Objectives

Flood Risk Prediction Analysis

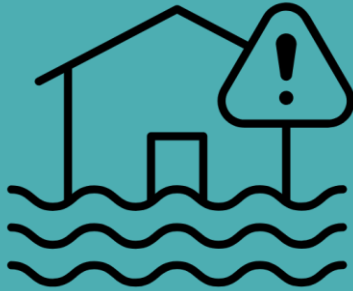


Image Credit: Savannah Vize

Implement a predictive flood model to assess the flood potential of unprotected wetland areas

Identify isolated freshwater wetlands in regions with high flood risk to highlight priority wetlands for conservation

Social Vulnerability Assessment



Image Credit: Ramakrishna Venkatesan

Create a bivariate map that showcases socio-environmental vulnerability crossed with flood risk predictions

Assess social vulnerability and flood risk of partner-identified communities of interest



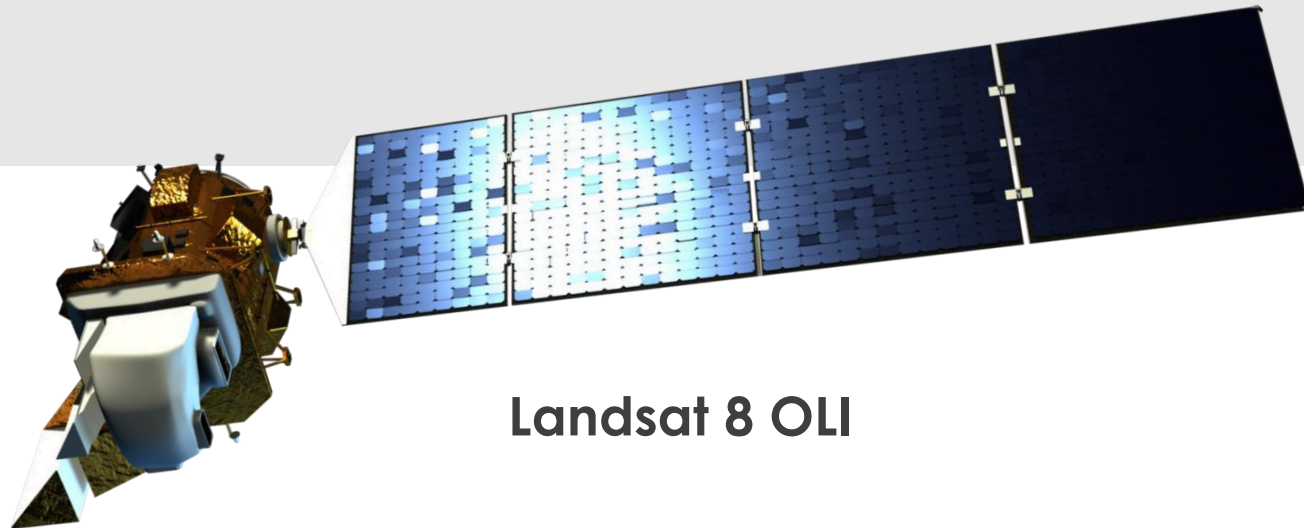
Earth Observations



Landsat 7 ETM+



Sentinel-1 C-SAR

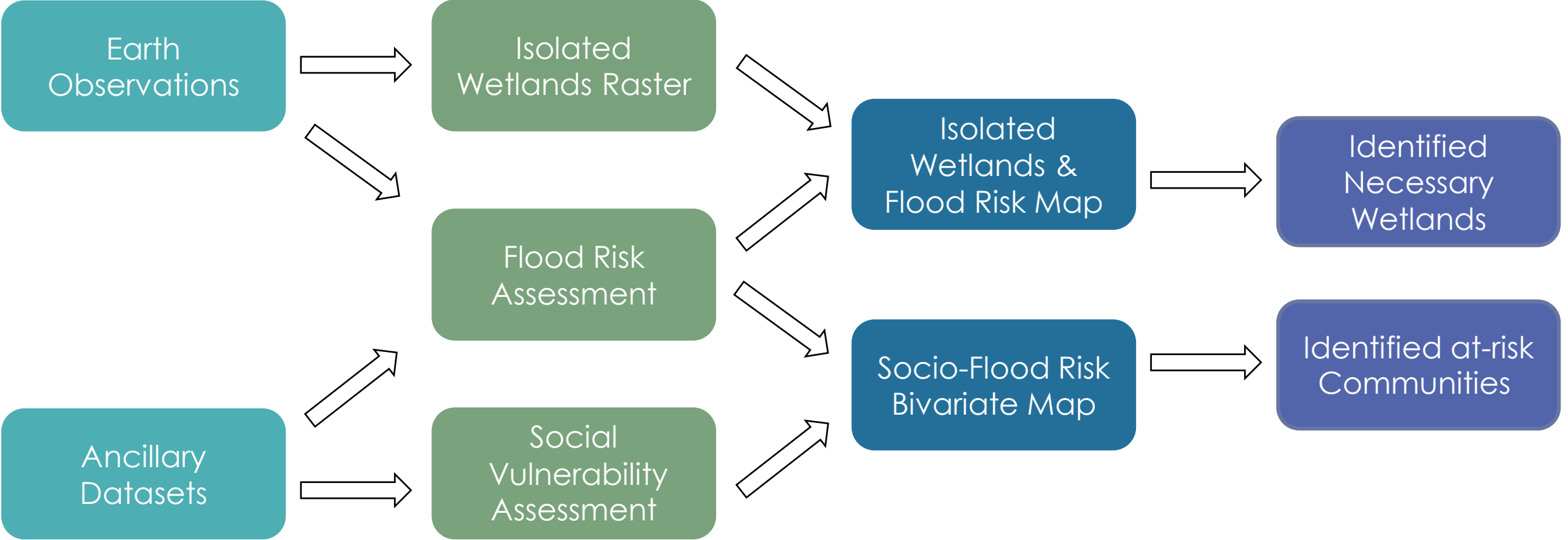


Landsat 8 OLI

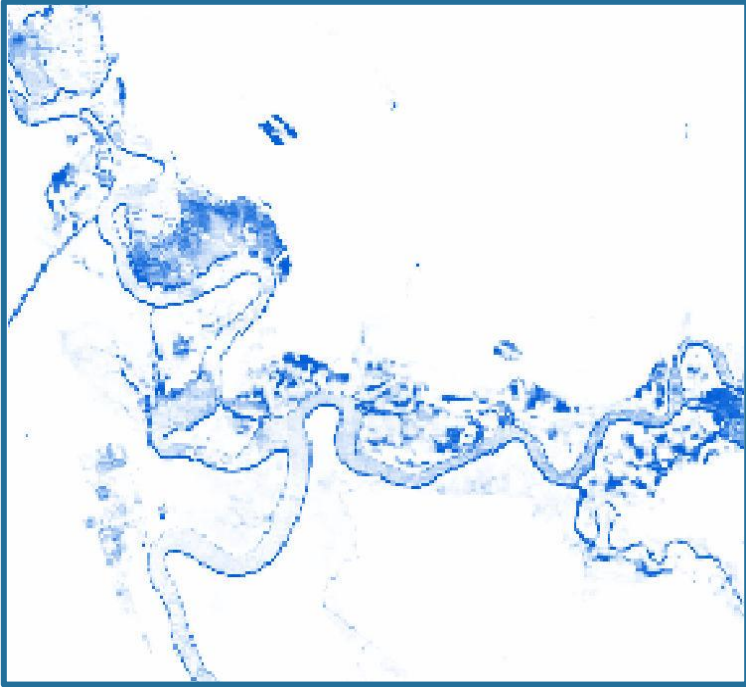
Image Credits: NASA, ESA



Methods – Objectives Flow Chart

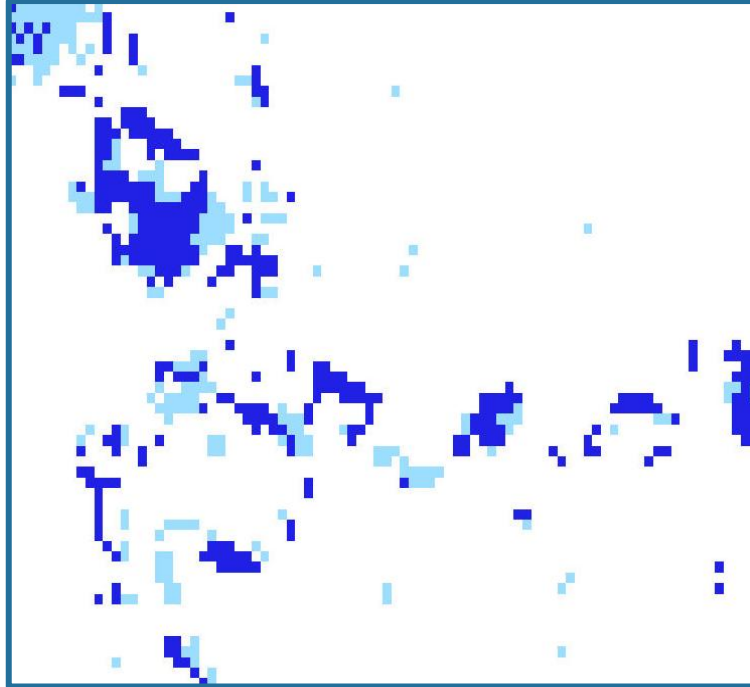


Methods – Mapping Flood Extent



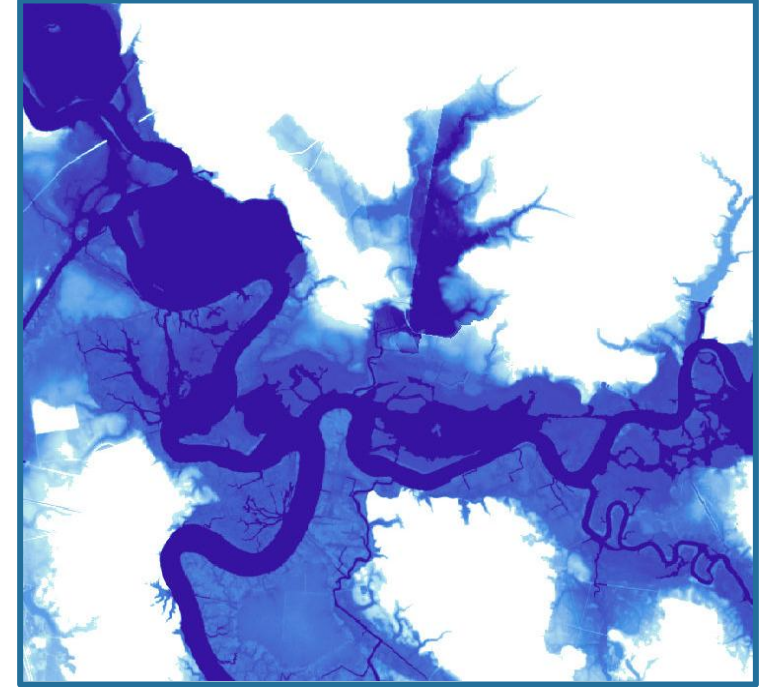
FLATS

*Landsat EVI + NDWI
Optical*



SAR

*VV Polarization
Radar*



HAND

*DEM + CGD
Model*

All imagery retrieved from Google Earth Engine



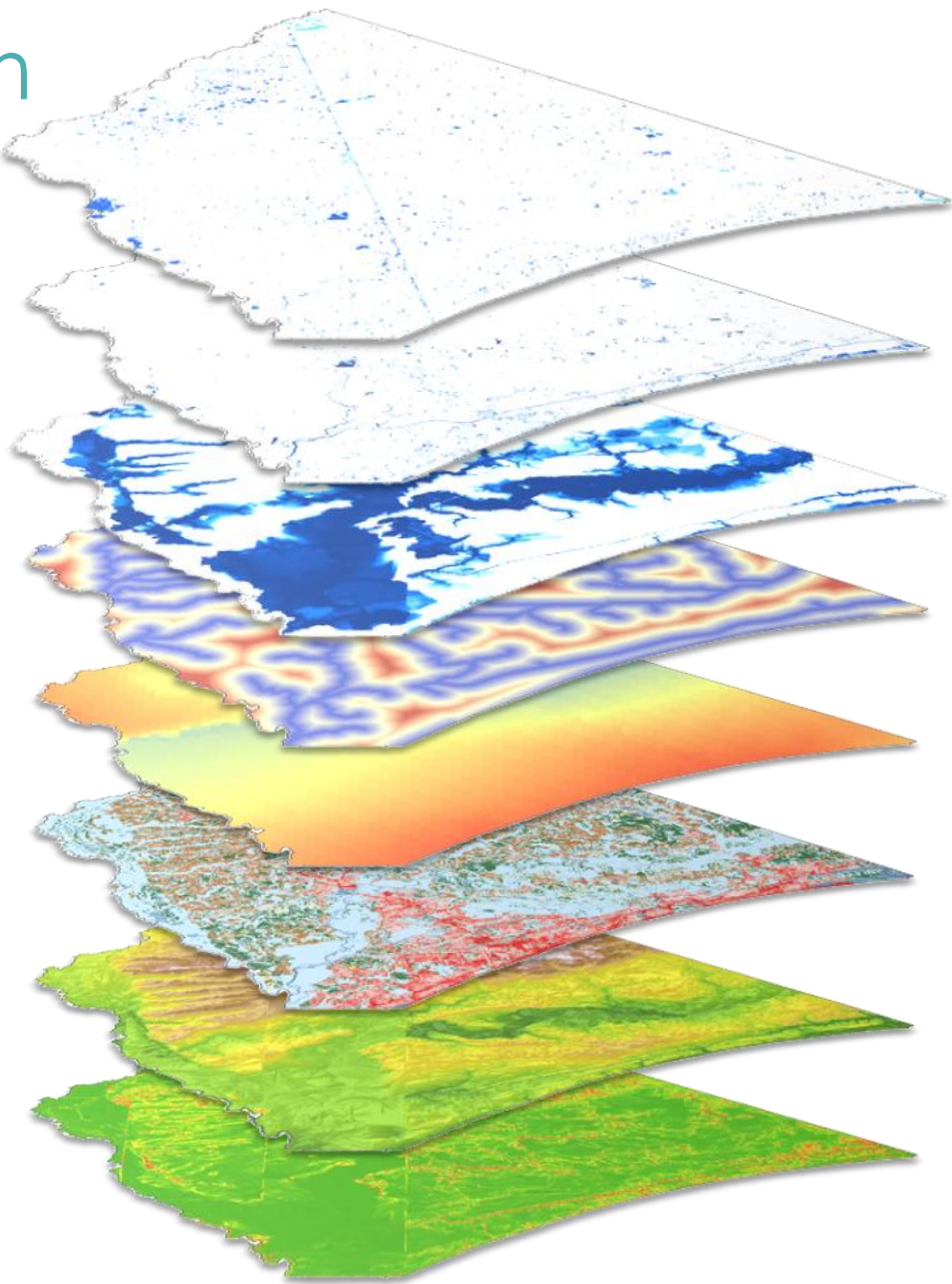
Methods – Flood Risk Weighted Sum

Flood Extents

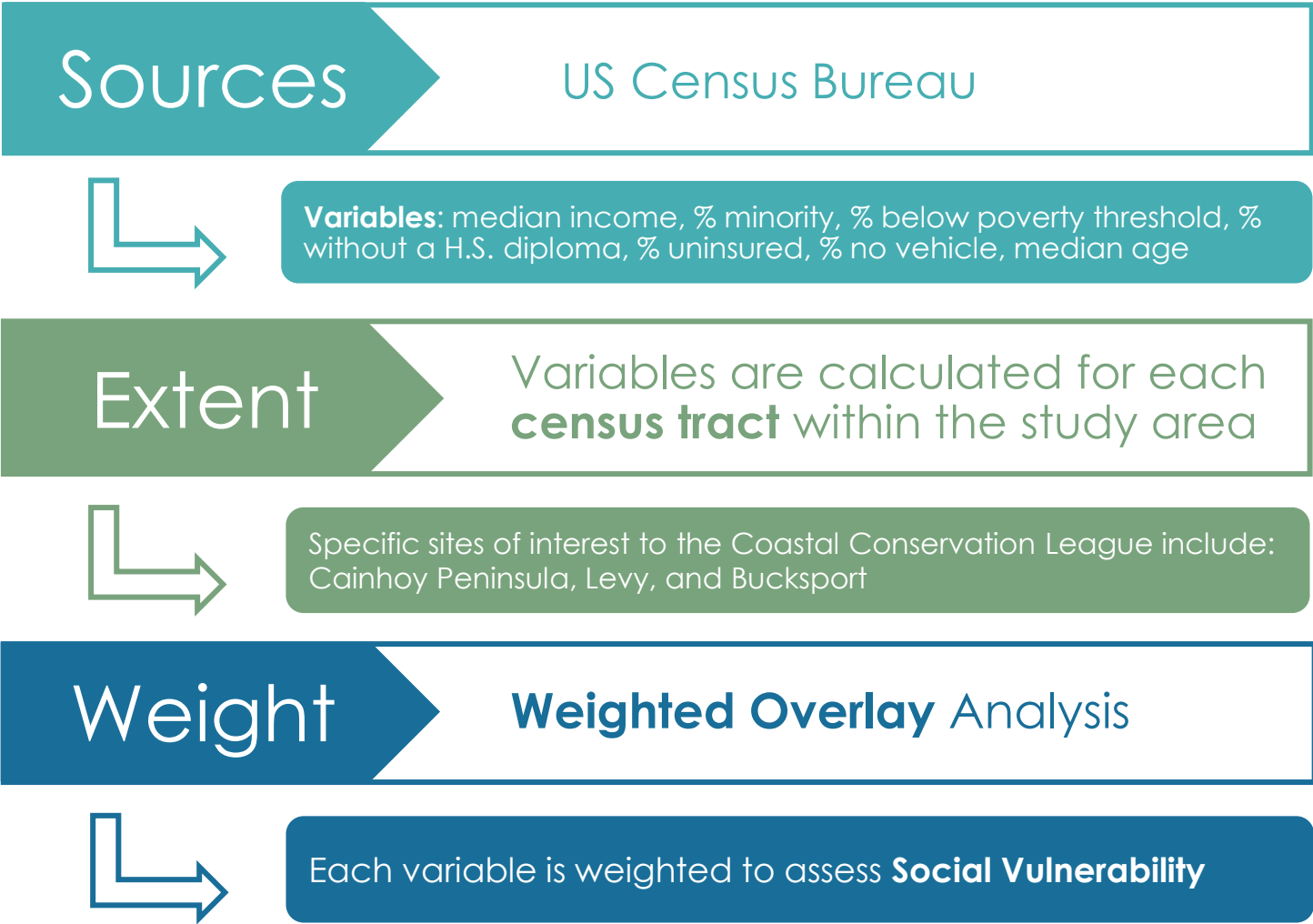
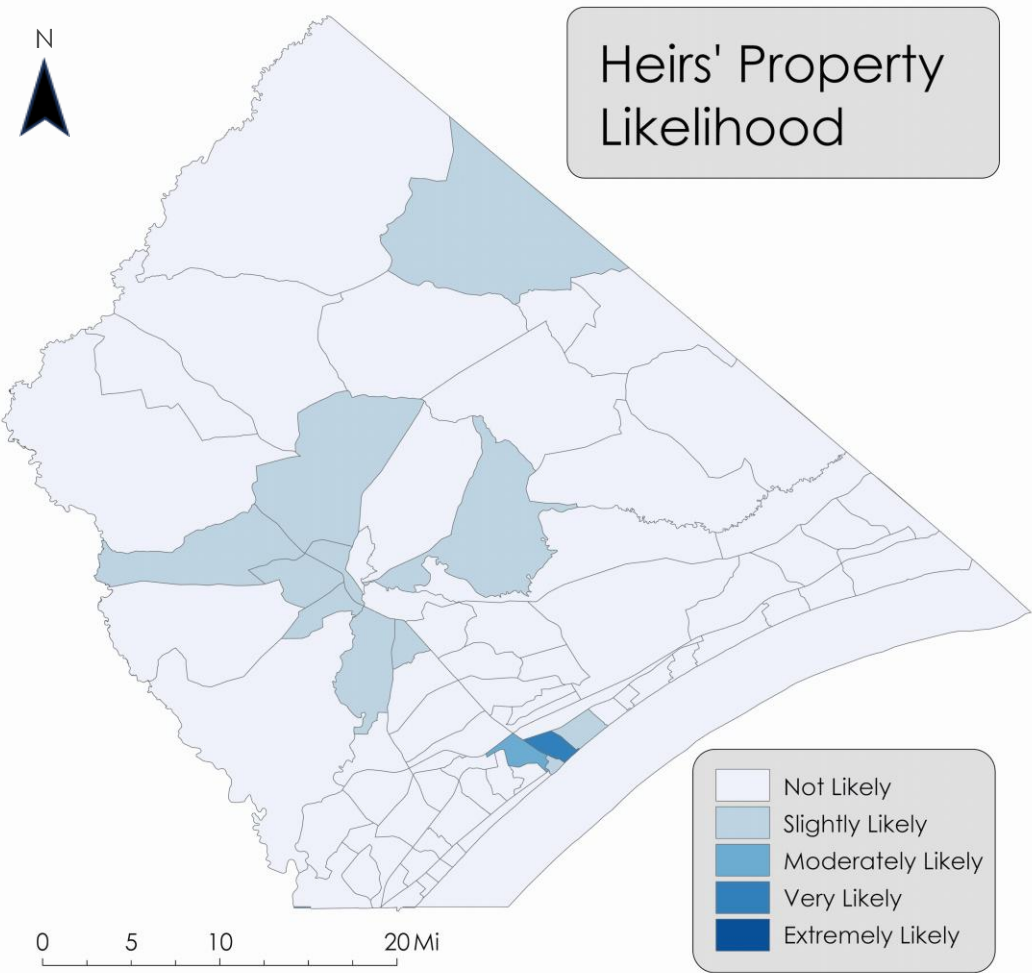
SAR Observation	20%
Landsat EVI and NDWI	10%
HAND Model	20%

Flood Predictors

Distance from Streams	20%
Precipitation*	0%*
Land Use Land Cover	10%
Elevation	10%
Slope	10%



Methods – Social Vulnerability Analysis



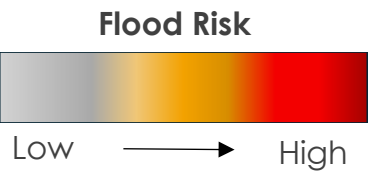
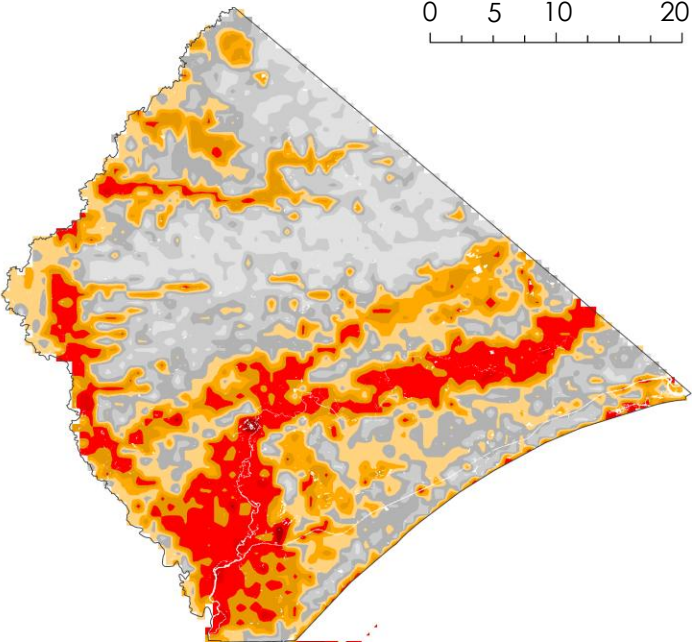
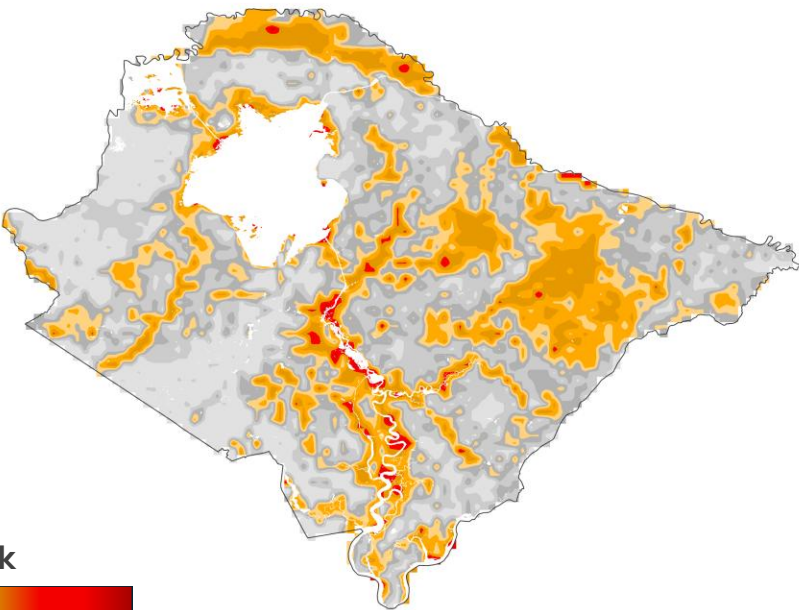
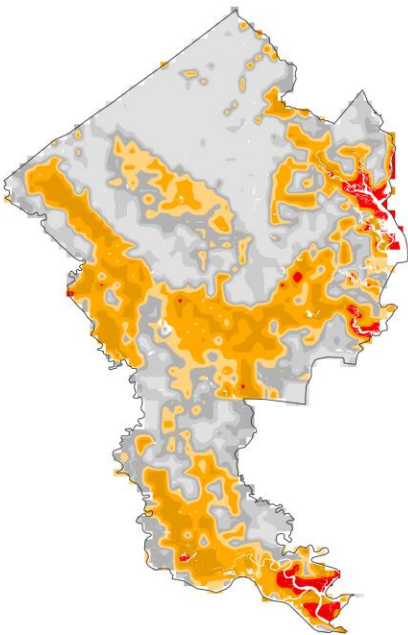
Results – Flood Risk Assessment

With Precipitation

Jasper

Berkeley

Horry



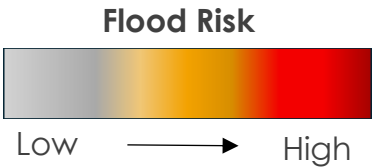
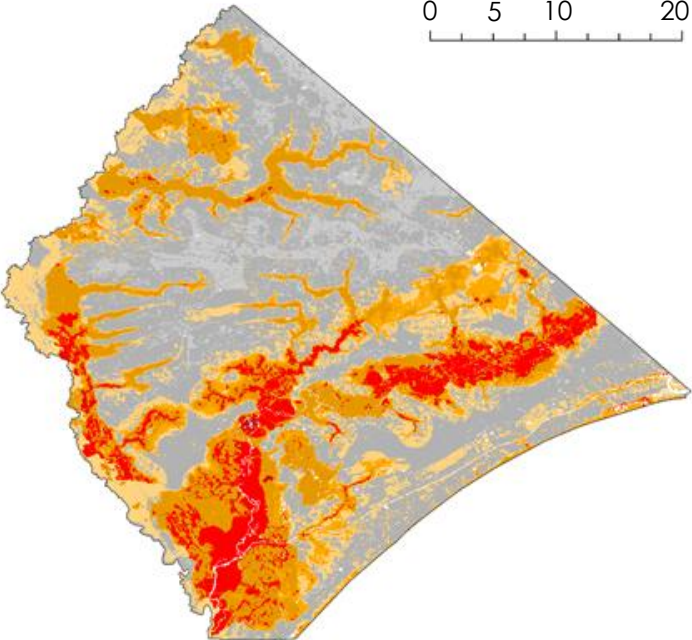
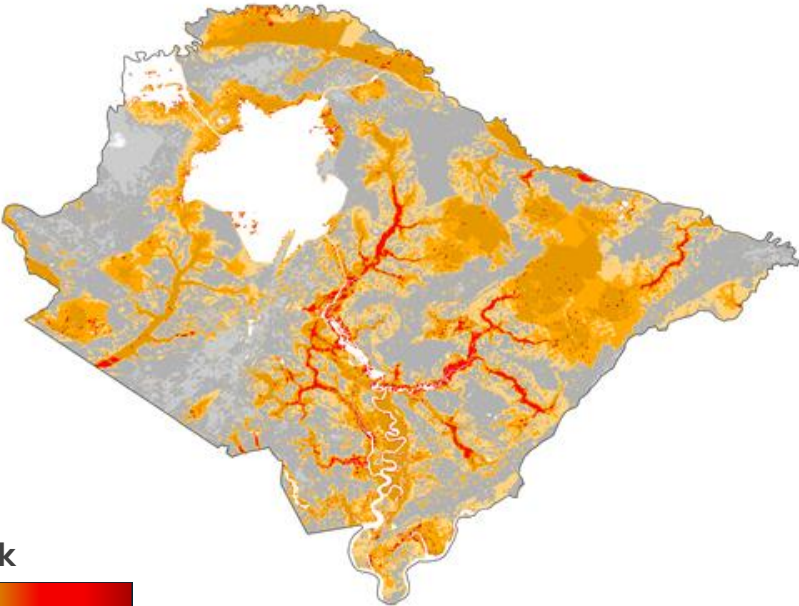
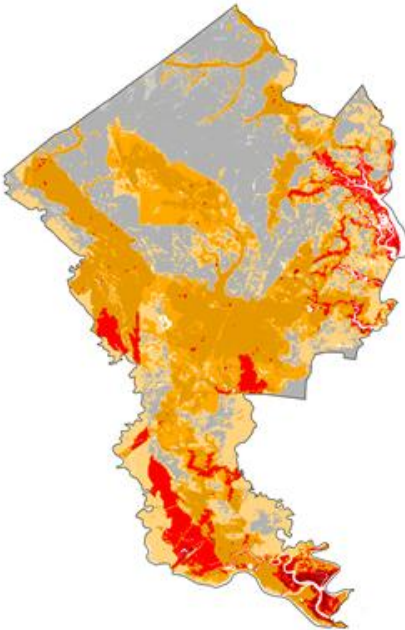
Results – Flood Risk Assessment

Without Precipitation

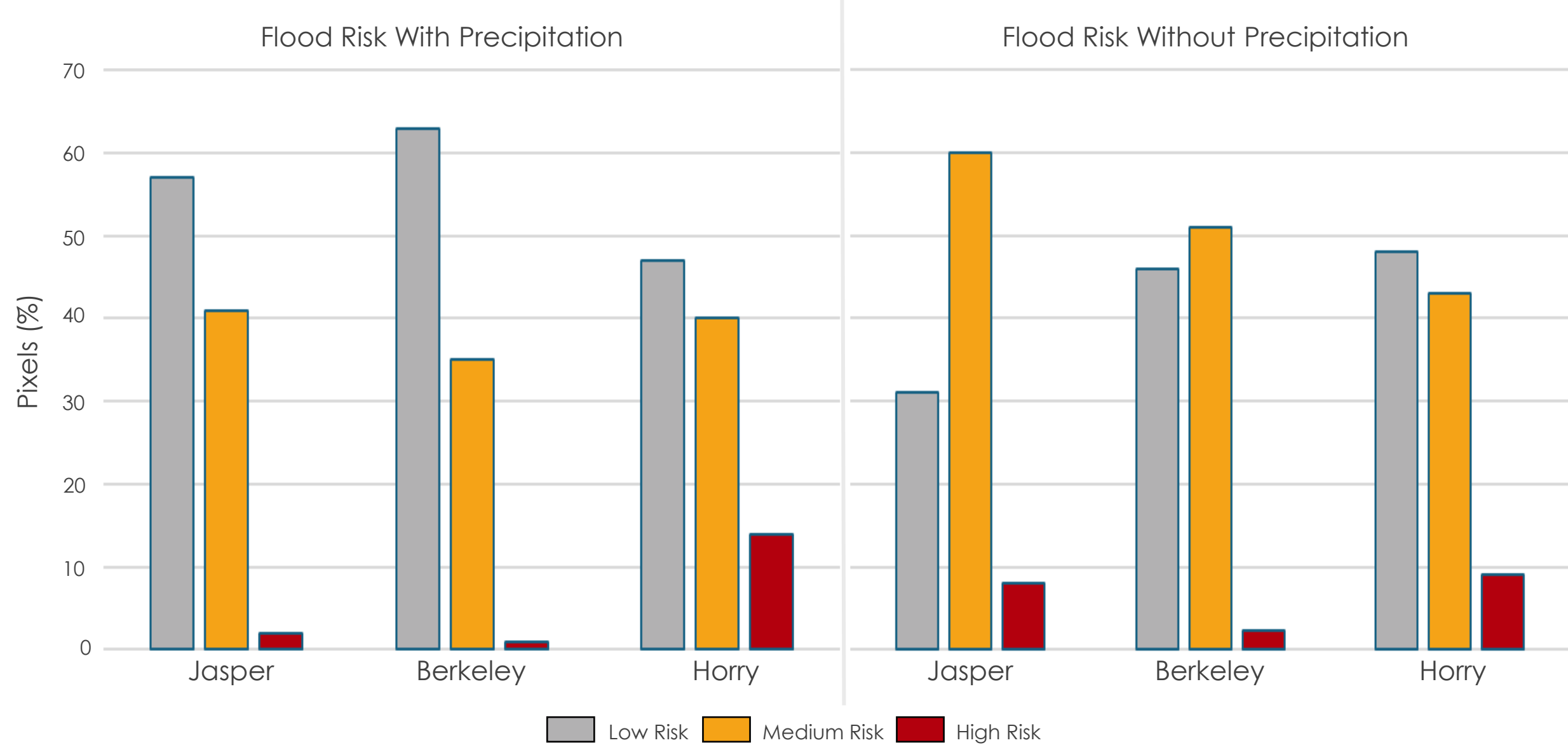
Jasper

Berkeley

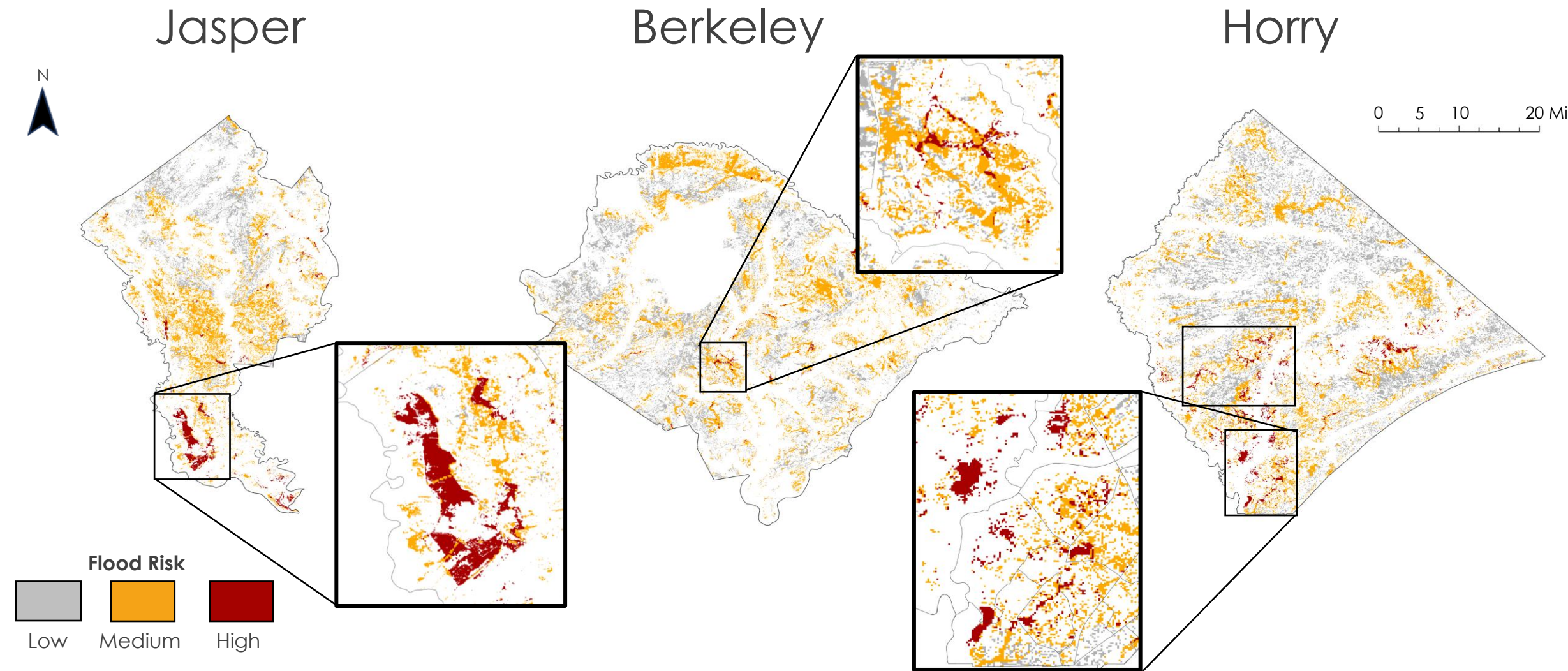
Horry



Results – Flood Risk Assessment Statistics

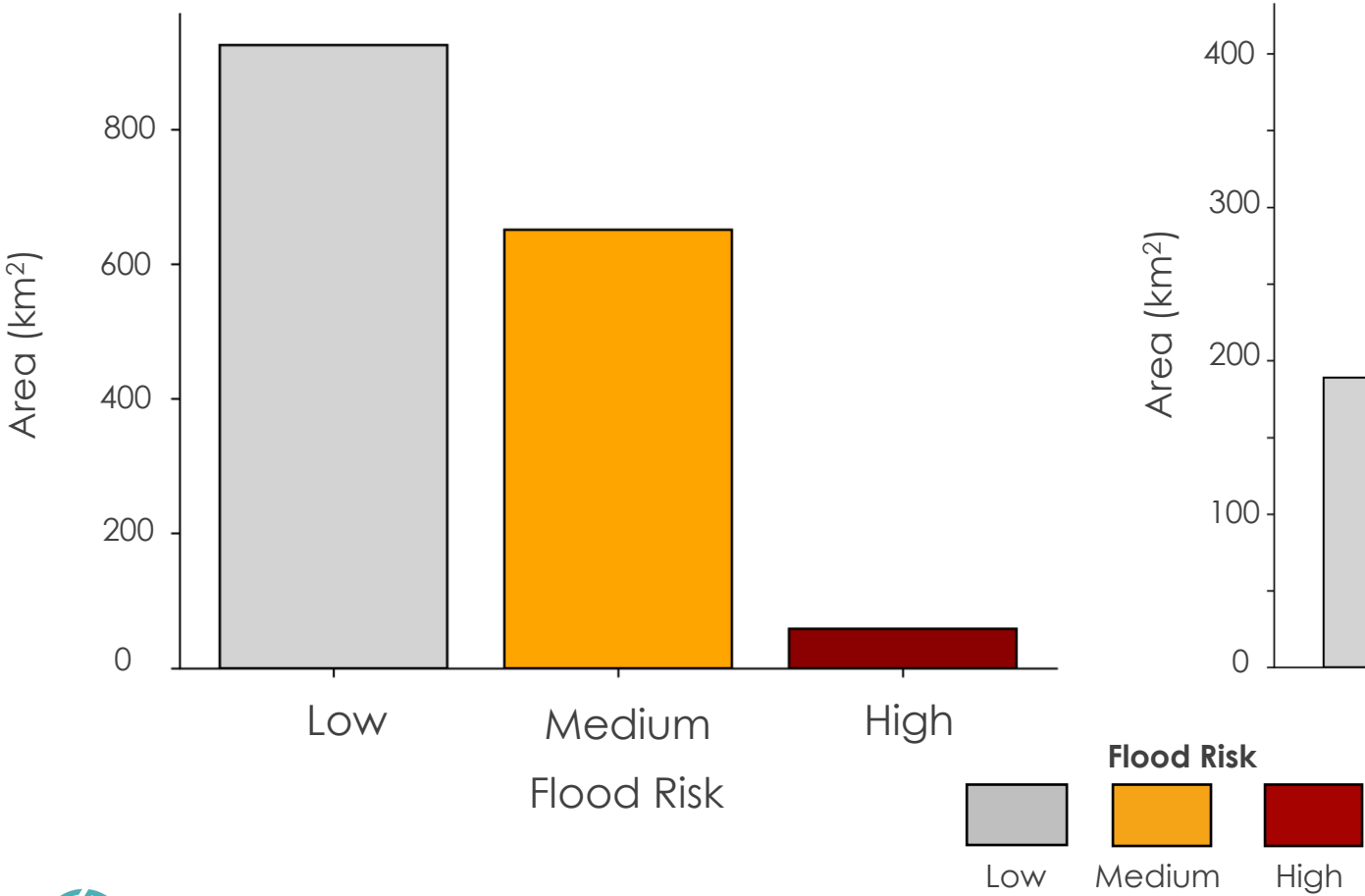


Results – Isolated Wetlands and Flood Risk

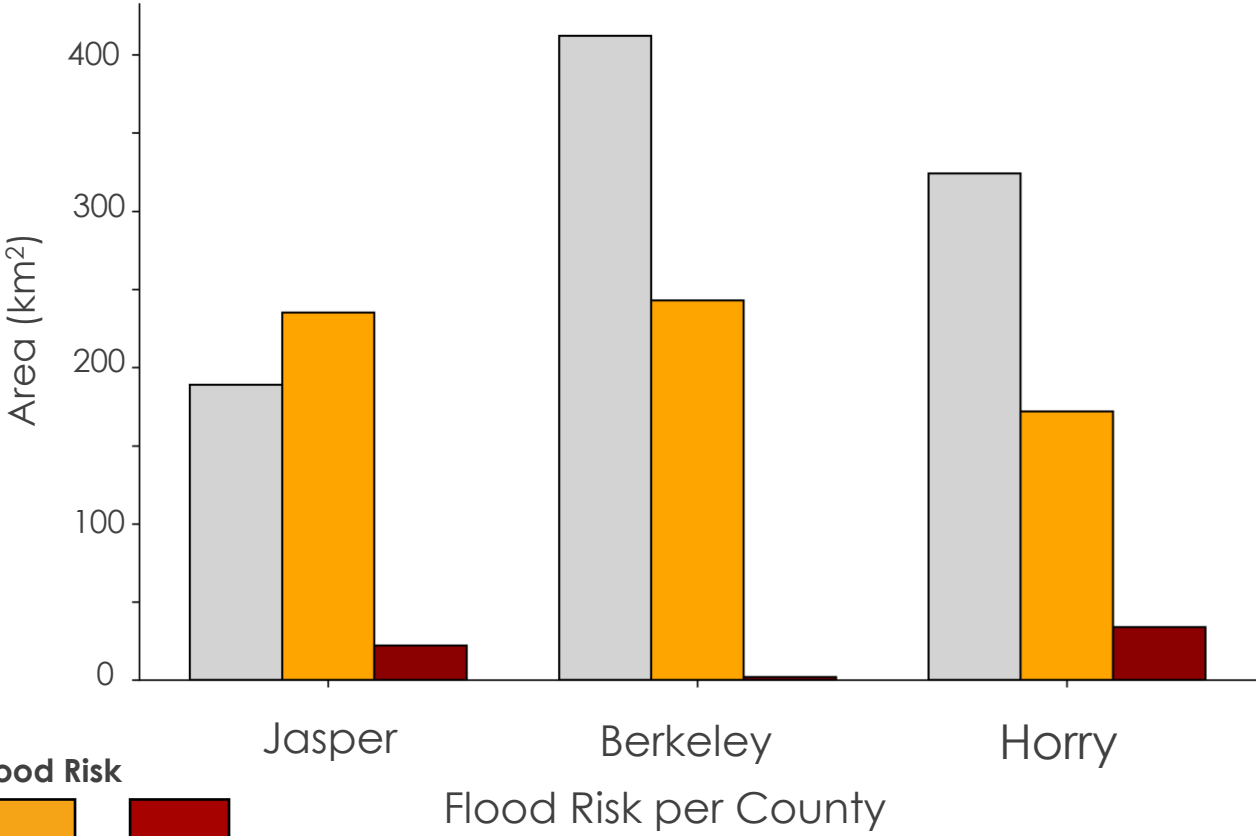


Results – Isolated Wetlands and Flood Risk Statistics

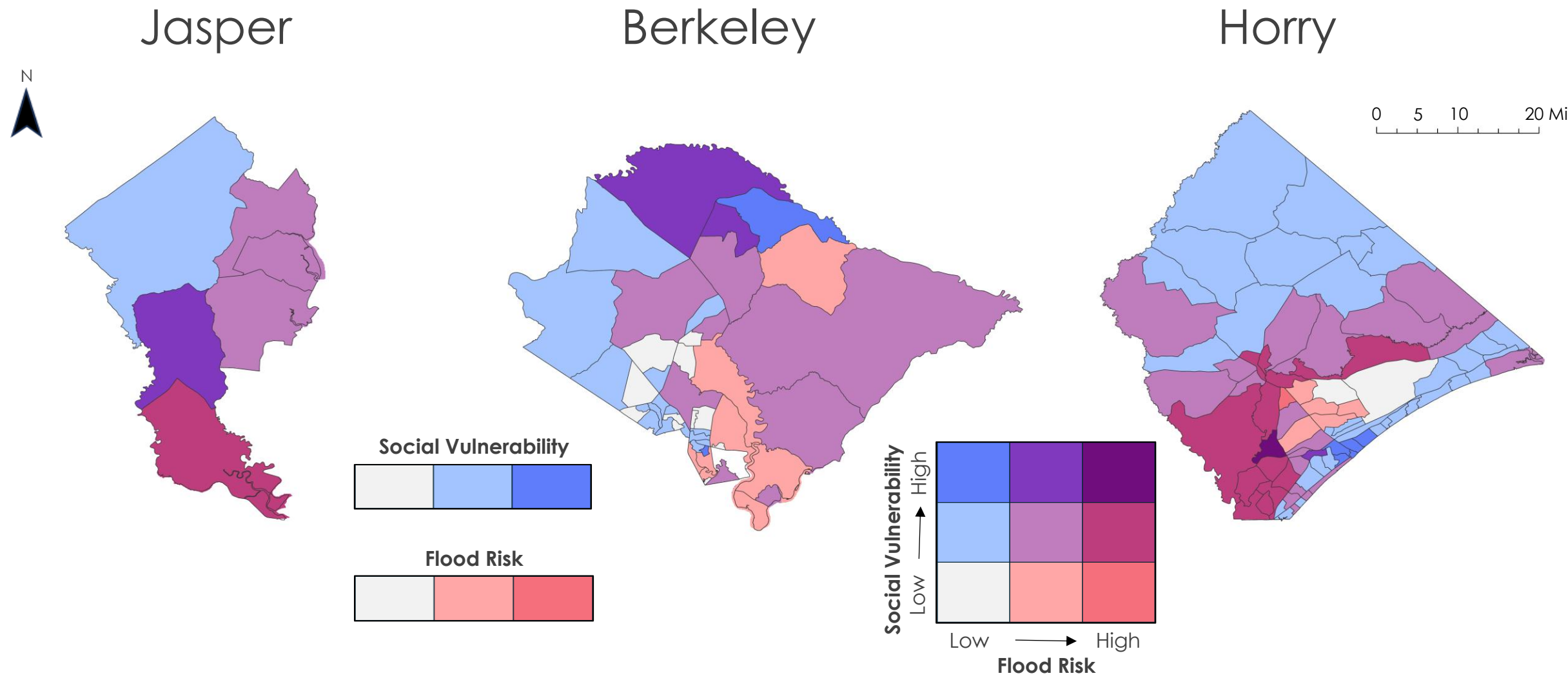
All Isolated Wetlands Combined



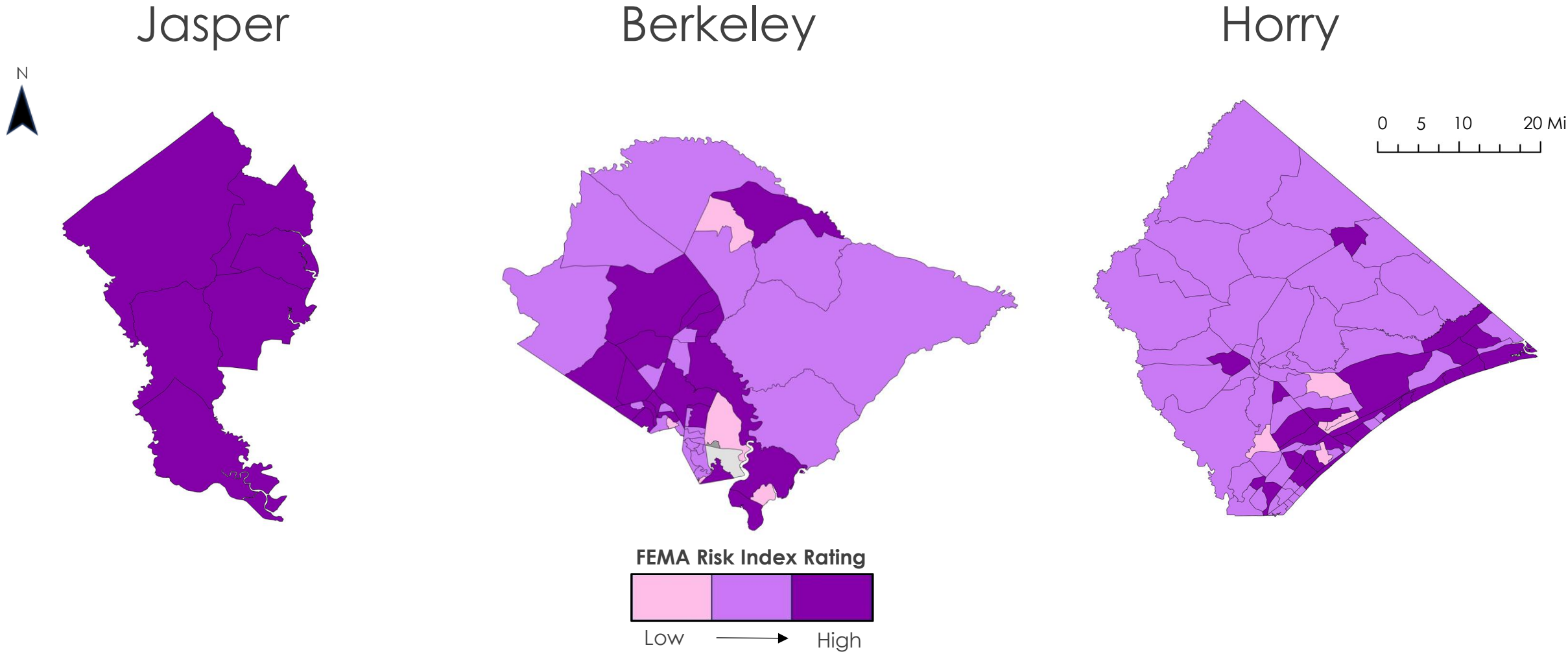
Isolated Wetlands by County



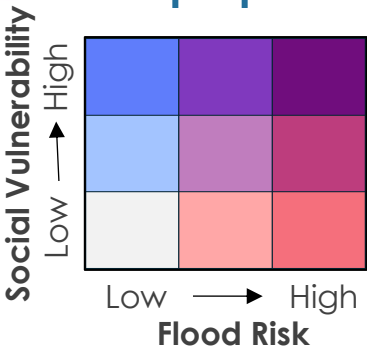
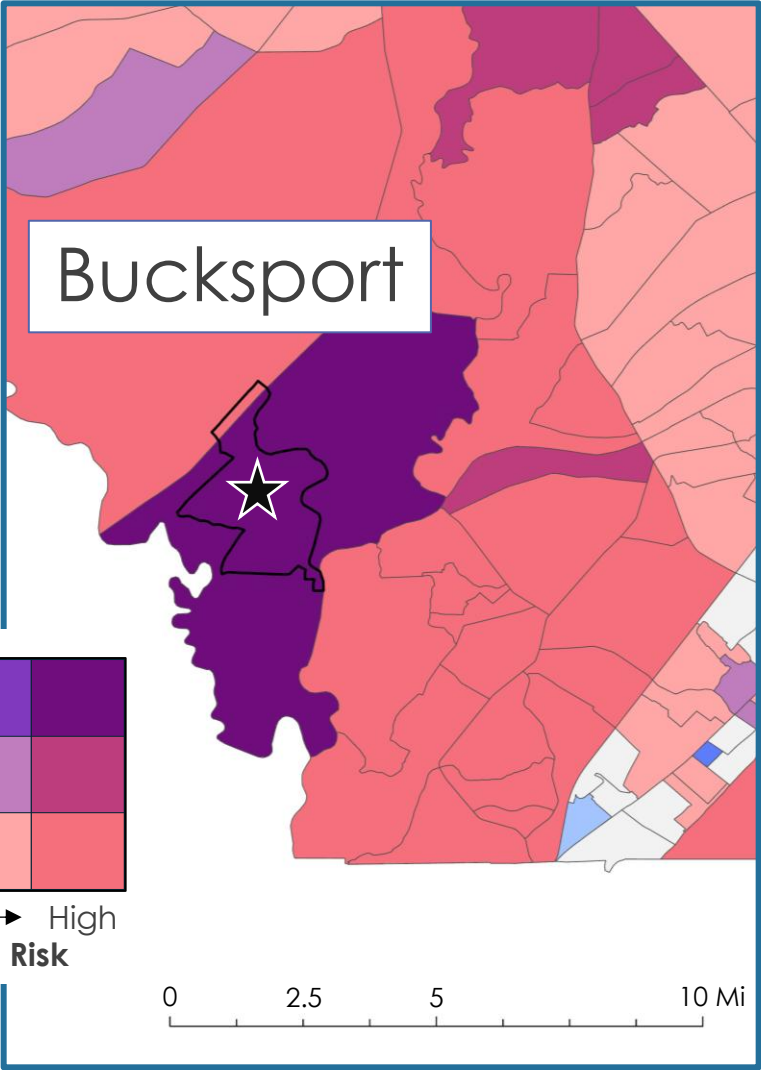
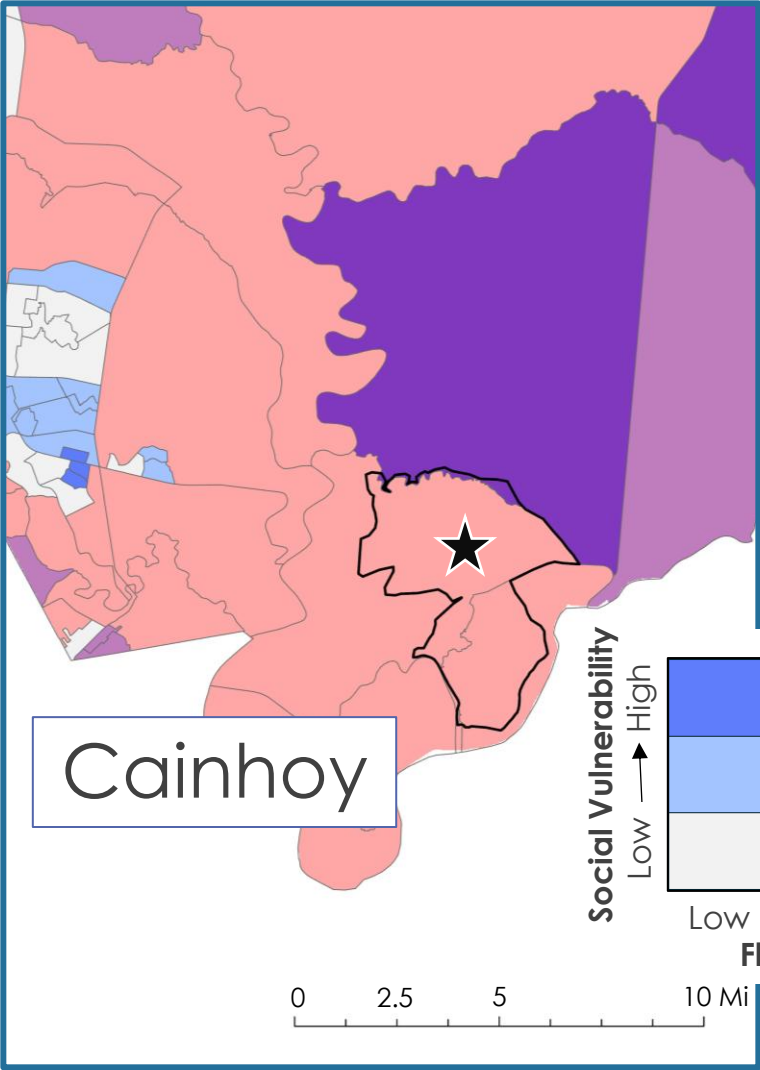
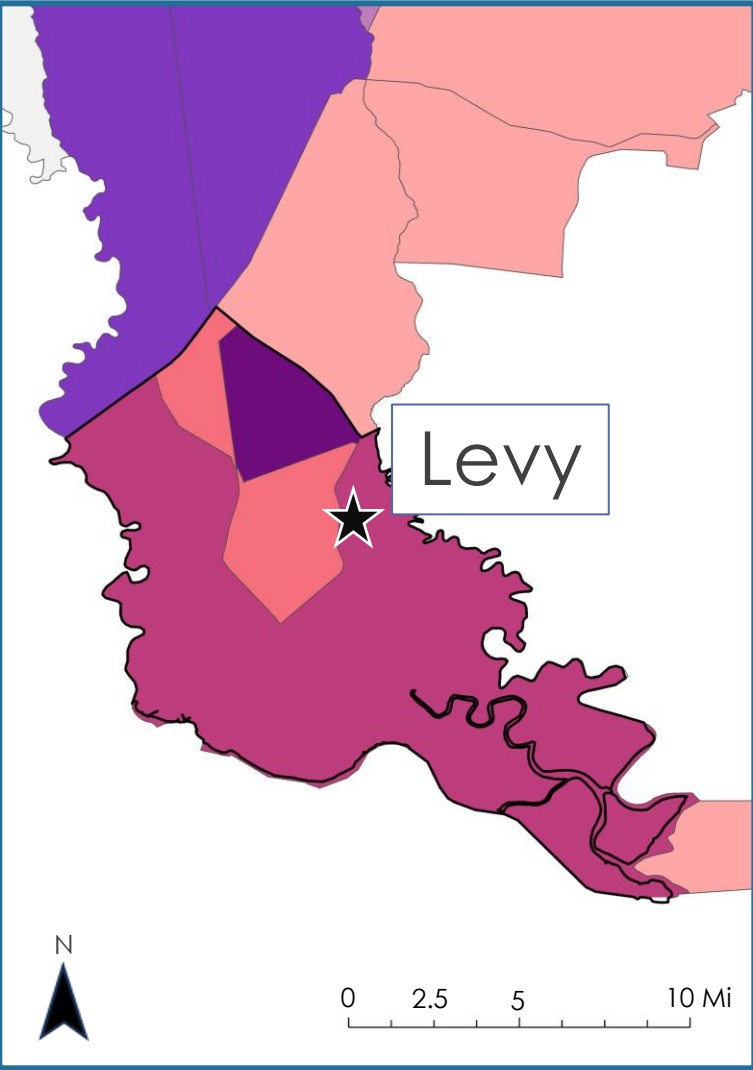
Results – Flood Risk and Social Vulnerability Bivariate Map



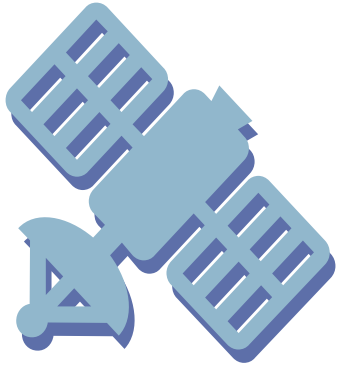
Results – FEMA National Risk Index Comparison



Results – Highlighted Communities



Errors and Uncertainties



Data Acquisition

- Lack of ground-truth data
- Gaps in satellite overpass during flooding event

Mapping Flood Extent

- HAND: relies solely on DEM data
- SAR: limited ability to detect forest flooding; backscattering

Scaling

- Subjective weighted scale
- Scale mismatch among demographic factors

Holistic Flood Risk

- Lack of multiple storm datasets
- Removing precipitation as a model factor



Feasibility – Partner Implementation

Earth Observations

Allow partners to assess flood and social vulnerability at broad scales

Partner Use

Partners will see at-risk regions where isolated wetlands should be protected

Future Application

Incorporate more data related to social vulnerability and wetland flooding



Conclusion

1

43% of isolated freshwater wetlands were in areas with medium-to-high risk flood zones. Two of the communities of interest, Levy and Bucksport, were located near isolated wetlands in high-risk flood zones.

2

Partner-identified communities of interest had medium-to-high flood risk and social vulnerability.

3

Combining optical and active radar data allows our partner to quickly prioritize wetland conservation in specific locations to address large-scale flooding resilience.



Acknowledgements

NASA DEVELOP

- Leads: Dr. Jennifer E. Mathis (Georgia - Athens) & Laramie Plott (Virginia – Langley)
- Science Advisor: Dr. Marguerite Madden (University of Georgia)

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DEVELOP Spring 2025 Team

- Jake Ferus, Yunjoo Cho, Sidney Eigeman, Maisunath Maliha Amin



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